REST API Testing — Day 1 Notes

1. What is a REST API?

Interview answer:

A REST API stands for Representational State Transfer Application Programming Interface. It allows two applications to communicate with each other over HTTP. It uses HTTP methods such as GET, POST, PUT, and DELETE to retrieve, create, update, and delete data.

Example: In a railway booking application, we can use a GET request to check seat availability without booking a ticket.

Interview Answer 2:

A REST API is an interface that allows different applications to communicate with each other over a network. It enables applications to send requests to a server and receive responses.

Example:

Consider a train ticket booking application. A user books a seat using a laptop, and another user opens the same application on a mobile phone. The application communicates with the server through APIs to show updated seat availability.

This helps ensure that a booked seat is no longer shown as available on other devices.

2. What is REST API testing?

Interview answer:

REST API testing is a type of software testing used to verify whether a REST API is working correctly. It involves sending HTTP requests such as GET, POST, PUT, and DELETE to the server and validating the responses.

We check the status code, response body, response time, and data accuracy to ensure the API works as expected.

Example:

In a railway ticket booking application, we can send a GET request to check seat availability. If the server returns a 200 OK status code and the correct seat details, we can verify that the API is working as expected

3. What are HTTP methods in REST APIs?

HTTP methods specify what action we want to perform on a resource.

There are four common HTTP methods you learned today:

GET

Purpose: Retrieve data from the server.

Example: Check whether seat A10 is available without booking it.

POST

Purpose: Create a resource or submit data for processing.

Example: Create a new ticket booking.

PUT

Purpose: Replace a resource with the supplied representation, usually updating an existing resource.

Example: Update a booking's details by replacing its existing data.

DELETE

Purpose: Delete a resource.

Example: Cancel a booking, if the API supports cancellation through DELETE.

Note: PUT is generally used to replace a resource. PATCH is another common method used when you want to update only specific fields.

4. How do you choose an HTTP method?

Interview answer:

We choose an HTTP method based on the action we want to perform on the resource.

- GET — retrieve information.
- POST — create or submit information.
- PUT — replace an existing resource.
- DELETE — remove a resource.

Example interview question: You want to check whether seat A10 is available without booking it. Which method would you use?

Answer: I would use the GET method because I only want to retrieve the seat availability information without creating a booking.

5. Quick revision

Question	**Key answer**
What is a REST API?	An interface for communication between applications.
What is REST API testing?	Verifying API functionality and responses.
Which method retrieves data?	GET
Which method creates a resource?	POST
Which method replaces a resource?	PUT
Which method deletes a resource?	DELETE

Day 02 Lesson:

Your next lesson: HTTP status codes such as 200, 201, 400, 401, 404, and 500. These are essential for REST API testing interviews.